

Forecasting the 2026 Hungarian Parliamentary Election

TDK presentation transcript

Start

Hi, my name is Kristóf and I want to show you how I predicted the 2026 Hungarian elections.

Motivation

First of all I would like to explain why this topic is important.

Predicting election outcomes can be useful in three ways:

1. First, political parties can use forecasts to see if their strategies are actually working.
2. Second, regular citizens can get an idea of how things might change politically in the future.
3. Third, researchers can use forecasts to understand voting behaviour. If we can predict something, maybe we can begin to understand and even explain why it is happening.

The Hungarian election was a pretty straightforward choice, since it has now happened. So I could not only build the forecast but also evaluate it against the real outcome. In this way I had to make a real world prediction from my model. I even pre-registered my predictions.

So the question I asked in my thesis is: how can we combine information from polls and other sources to make a good prediction about elections?

What others did

Moving on to how others usually do a prediction.

First let me talk about the inputs. Most election forecast models use polls, and when polls are done right they are a direct signal of public opinion.

Forecasters typically aggregate polls, filtering or weighting them based on past performance.

Some models add other data sources too. They build a structural baseline forecast using demographics or economic indicators.

Then the model transforms the input data into vote share forecasts. After that it must transform these forecasts into projections about parliamentary seats. This is usually done by simulation, to obtain estimates not only of which party will win how many seats but also of the uncertainty of the model.

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Model used: anthropic Claude Opus 4.7

I followed this approach with one extension. I also used media consumption data, because Hungary has a special media landscape where many outlets are described as government aligned, so their geographic viewership could carry a meaningful political signal.

This is my prediction pipeline, where I could cover the last 2 national elections and the last 2 European Parliament elections. Now let me talk more about the data I used.

Data sources

The election results and polls come from Prof. Gábor Tóka's website, which has been collecting polls since 2014.

Local and national economic indicators come from KSH.

Online media usage comes from Gemius, an audience measurement company. The data covered several outlets like telex, index, 24.hu, RTL and TV2.

I aggregated the pro-government or government aligned outlets. I used ATLATSZO.HU's data, which summarizes which outlets are financed by the government, to categorize outlets into pro-gov and opposition. So for objective reasons I assumed that outlets financed by the government are politically aligned with it.

I observed media use intensity (average user) of the pro-government and opposition platforms in the 2 months before the election.

Now I will give a high level overview of the model.

The model and the results

From the polls and from the demographics we get a national list share prediction. Then from the district lean we get the district level list shares, and from the candidate gap we get the votes for candidates and finally the seats.

Before moving on I want to mention that everything that needed to be tested or tuned was tested using Leave One Election Out (LOEO). I left one election out of the entire pipeline and then tested on it.

Poll model

First, here is how I used polling data.

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I used a simple weighted average, using the sample sizes of the pollsters as weights. This is easy to interpret.

On top of that there is a second more advanced model. It corrects for house effects, which means that certain pollsters tend to give larger or smaller estimates for certain blocs or parties. The model learns this from past elections and corrects for it.

After this a so called Kalman filter is applied. The filter assumes public opinion changes slightly each day. When there is no poll for a day it carries the previous estimate forward with a bit of added uncertainty. So a random walk. When polls are less reliable it sticks closer to the previous day rather than fully relying on the unreliable poll.

One really important thing. Since political parties were always changing, I grouped parties into blocs. So in past elections I can use these blocs to train and test, instead of always changing parties. This way I could estimate the opposition's past performance.

Here are the results. These are basically the 2026 prediction for list votes. The left one is the simple poll average, while the right one has the house effects plus random walk applied. It can be seen that the right one is smoother than the raw average.

In the simple average the opposition held an almost clear lead throughout the observed window, while in the Kalman filter it was not that obvious. There were ups and downs. So the question was: which to choose? It was a difficult question.

The house effects model is theoretically better but needs more historical elections to estimate pollster biases reliably. With only 4 elections available it picks up noise instead of real signal. The simple average won 3 backtests out of 4. So it is more trustworthy here, given the data we have.

Structural baseline

Moving on to the next part, I would like to show you how I used demographic and economic data.

The idea is that these variables move slowly, so they give a good anchor next to the polls.

First I did a quick feature selection on past elections. The demographic variables turned out to be the most useful: log population, share of young population and share of old population. Other variables like the town hall's budget balance or doctors per capita made the historical errors worse.

So I used these variables to predict the district level vote shares with a Ridge regression, so I could penalize the coefficients.

This gives a predicted vote share for each district based on demographics alone, which I then aggregated to the national level to get a national list prediction.

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70-30 combination and district forecast

To summarize what happened so far. From the polls I got a national list share prediction. I also used the fundamentals (the demographics) to predict the national list share.

Polls can swing a lot in the short term or be biased, while the fundamentals prior can act as a structural anchor that pulls the forecast back. I then combined the polls with this prior using a 70 percent polls and 30 percent fundamentals rule, inspired by the literature.

From this national level list share I could forecast the individual district list shares and the individual district candidate votes. For these I used the district lean model and the candidate gap model. Since these are complicated models, I will focus only on the key ideas instead of going through every component.

District lean

I start with the district lean. Here I calculate the difference between the actual national list share and the district level result for each district. This difference is called the district lean. It is explained in 2 layers.

On the first level I explained this difference with the demographic variables I mentioned earlier.

In the second layer I subtracted the predicted lean from the real result, which gives a residual. So the first model predicts something and I explain the remaining difference again. I explained this remainder with media usage and past results (since these are limited, available only for the last 2 elections). This is a kind of recent correction layer.

So now we have a prediction for the district lean, namely how much each district will deviate from the national list share.

Candidate gap

In the district lean we examined the difference between national list share and district list share. In the candidate gap I examine the difference between the district list share and the district candidate share. In the Hungarian system, a bloc's candidate vote in a specific district is usually not equal to the bloc's list share in the same district.

I explained this difference with past gaps, the same demographic variables and the district list share as an anchor.

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Okay, so now we know everything: how to predict list share from polls and from fundamentals, how to get the district level list share from the predicted difference and how to get the candidate shares by district.

Seat translation and Monte Carlo

What is needed now is just to translate these predictions into seats. As you might know, there are 106 single-member district seats and 93 national-list seats. The compensation votes are important here in the Hungarian system. If a bloc loses a district, all of its candidate votes are added to its national list as compensation. If a bloc wins a district, only the surplus votes are added.

Then a so called D'Hondt rule is used to get the final seats, which is a popular proportional allocation rule used in many European systems.

But we need more than just a simple point estimate, and the last step to get this is a Monte Carlo simulation, which adds uncertainty to the predictions.

After each point prediction it draws random errors. At the district level the sample is large, so I sample errors directly from past residuals. At the national and candidate gap levels the sample is small, so I build a covariance matrix from past errors but pull the correlations toward zero to a certain extent before drawing, to avoid learning fake patterns. I did 1,500 draws and from this I could create a probability structure like this.

Results

These are my pre-registered forecasts, along with the point estimates on the next slide.

It can be seen that the combined forecast gave a higher probability to TISZA finishing first with a higher list share. However it gave a higher probability to a Fidesz majority in seats.

On the day before the election I chose this as my final model, because in the last parliamentary election and the last EP election it had performed better. However now we know that the simple poll aggregates were closer.

The simple poll only model gave a much higher probability to a TISZA list share win and to a TISZA majority. In the seat distributions it gave a clear separation for a TISZA majority. But it was still not close enough to the real result, which was 141 seats for TISZA.

However, looking only at the list results, these predictions were much closer than the overall seat predictions.

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Path	Government	Opposition (TISZA)	Radical opp. (Mi Haz.)	Other
Simple weighted average (poll only)	40.1	48.7	5.6	5.6
Real result	38.61	53.18	5.63	2.58

Overall, what I found interesting about the results is that the simple poll averages were closer to the truth than the combined forecast. The simple poll only model got the direction right but the magnitude was wrong. There are a few clear reasons for this.

1. **This was a sea change election.** The 2026 election broke the patterns of the previous four cycles. The structural prior was the main source of error. It was trained on four Fidesz era cycles and carried a Fidesz era political geography into 2026, where that geography no longer fits. The district leans were also trained on this data, which is why the SMD predictions were much less accurate than the list share predictions. On top of this, turnout reached a record level. Many previously inactive voters showed up and disproportionately voted for TISZA, a shift that no model trained on historical turnout patterns can capture in advance.

The reviewers asked me to generalize this beyond Hungary, so I would say: any prior trained on a stable period will fail when voting behaviour shifts in ways the data cannot capture. The lesson for choosing between methods is simple. When the polls and the prior disagree more than usual, we should give the polls more weight, because polls capture better what is happening now while the prior is built on the past.

2. **Polling quality.** Some pollsters underestimated TISZA significantly. Based on past backtesting and the literature the model should not be this far off, which raises a real question about polling trustworthiness. So there is a suspicion that some pollsters were aiming to influence public opinion rather than measure it accurately.
3. **Garbage in, garbage out.** This is an old rule of data science. Even a well validated model cannot recover from systematically biased input. The model's structure was sound on backtests, but in 2026 both inputs (polls and historical patterns) carried errors that the pipeline could not correct for.

Discussion

So about where this leaves us.

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Going back to the question I started with: how can we combine information from polls and other sources to make a good prediction about elections?

Based on this exercise, the answer depends on the political context. In stable periods, combining polls with a structural prior helps. The combined forecast was on average better across the four backtests. But in 2026 the context was shifting and the simple poll average alone got the direction of the result right. So combining works when the underlying patterns are stable. When they are not, the polls should stand alone.

Other sources like demographics, economic indicators and media consumption do influence voting, but in complex and indirect ways that are hard to capture in a model trained on only four cycles. They can serve as a structural anchor that stops the forecast from overreacting to short term polling swings, but you have to be careful with them. In 2026 the structural prior actually pulled the combined forecast away from the truth, because it was trained on Fidesz era political geography that no longer fit.

There are two clear directions for future work. First, with more election cycles the house effect estimates will become more reliable and the structural prior can be tested across more than one regime. Second, adding a turnout layer would help catch mobilization shocks that demographics alone cannot predict.

I will close with a quote that I think captures the spirit of this exercise.

"Prediction is very difficult, especially about the future." — Niels Bohr
(attributed)